

3. Formula reference

Notation: $V_{LL} = 208\text{ V}$, $V_{LN} = 120\text{ V}$, $\sqrt{3} \times 208 = 360.3$. P in W or kW, S in VA or kVA, Q in var, L = one-way length in feet, CM = circular mils, K = $12.9\ \Omega\cdot\text{cmil/ft}$ (Cu), 21.2 (Al).

3.1 Load current and power

1Ø L-N (120 V): $I = P / (120 \times \text{PF})$ $S = 120 \times I$
1Ø L-L (208 V): $I = P / (208 \times \text{PF})$ $S = 208 \times I$
3Ø (208 V): $I = P / (360.3 \times \text{PF})$ $S = 360.3 \times I$
Reactive: $Q = P \times \tan(\cos^{-1} \text{PF})$ $Q = \sqrt{S^2 - P^2}$
kW from kVA: $P = S \times \text{PF}$
Motors: use NEC Table 430.248 (1Ø) / 430.250 (3Ø) FLC, not nameplate [430.6(A)(1)]
Packaged HVAC: use nameplate MCA and MOP [440.4(B)]
General receptacle outlets: 180 VA per single or multiple yoke [220.14(I)]

3.2 Per-phase loading and balance

L-N load → adds I to its phase
L-L load → adds full I to BOTH phases it spans (arithmetic, conservative)
3Ø load → adds I to all three phases

$I_A = \Sigma(\text{L-N on A}) + \Sigma(\text{L-L on A-B and C-A}) + \Sigma(3Ø)$ (same for B, C)
 $I_{\text{avg}} = (I_A + I_B + I_C) / 3$
Current imbalance % = $(I_{\text{max}} - I_{\text{avg}}) / I_{\text{avg}} \times 100$ design target $\leq 10\%$
Voltage unbalance % = $\max |V_x - V_{\text{avg}}| / V_{\text{avg}} \times 100$ (NEMA MG 1: derate motors > 1%)
Motor derating (MG 1): approx. factor from MG 1 Fig. 14-1 (≈ 0.90 at 3%)

3.3 Busway / feeder totals (vector)

$P_{\text{total}} = \Sigma P$ $Q_{\text{total}} = \Sigma Q$
 $S_{\text{total}} = \sqrt{P_{\text{total}}^2 + Q_{\text{total}}^2}$
 $\text{PF}_{\text{total}} = P_{\text{total}} / S_{\text{total}}$
 $I_{\text{equivalent}} = S_{\text{total}} / 360.3$

3.4 Continuous-load sizing and OCP

Required ampacity & OCP $\geq 1.25 \times I_{\text{continuous}} + 1.00 \times I_{\text{noncontinuous}}$
branch circuits: 210.19(A)(1), 210.20(A) feeders: 215.2(A)(1), 215.3
100%-rated breaker AND enclosure listed for it → factor 1.00
Busway: rating $\geq$ design current and protected at its ampacity [368.17(A)]
Conductor ampacity $\geq$ OCP rating [240.4]
next standard size up allowed $\leq 800\text{ A}$ if not a multi-outlet
receptacle branch circuit [240.4(B)]
Small conductors: #14 $\leq 15\text{ A}$, #12 $\leq 20\text{ A}$, #10 $\leq 30\text{ A}$ (Cu) [240.4(D)]
Standard OCP sizes [240.6(A)]

3.5 Conductor ampacity — correction and adjustment

$I_{\text{allowable}} = I_{\text{table}} \times F_{\text{temp}} \times F_{\text{adj}}$
 F_{temp} : ambient correction, Table 310.15(B)(1) (e.g. 90°C insulation @ 40°C → 0.91)
 F_{adj} : >3 current-carrying conductors, Table 310.15(C)(1)
4–6 → 0.80 7–9 → 0.70 10–20 → 0.50
Derate from the 90°C column, then final value $\leq 75^\circ\text{C}$ column (terminal limit) [110.14(C)]
Counts as current-carrying: neutral of 3Ø 4W wye with major nonlinear load [310.15(E)(3)]
Does NOT count: EGC, IG [310.15(F)]

3.6 Voltage drop and minimum conductor size

3Ø: $VD = \sqrt{3} \times K \times I \times L / CM$ 1Ø: $VD = 2 \times K \times I \times L / CM$
Impedance method (more accurate, uses Ch. 9 Table 9, steel-conduit column for RGS):
 $VD_{3Ø} = \sqrt{3} \times I \times L \times (R \cos\theta + X \sin\theta) / 1000$ R, X in $\Omega/1000$ ft
 $VD_{1Ø} = 2 \times I \times L \times (R \cos\theta + X \sin\theta) / 1000$
Required CM: $CM_{min} = \sqrt{3} \times K \times I \times L / VD_{allowed}$ (3Ø)
Busway with evenly distributed load: use $L_{eff} \approx L / 2$ with Starline R, X data
 $VD\% = VD / V \times 100$ targets: 3% feeder, 5% feeder + branch [215.2(A)(1) IN 2, 210.19(A) IN 4]
Upsized for VD → upsize EGC: $CM_{EGC} = CM_{EGC,min} \times (CM_{phase,actual} / CM_{phase,min})$ [250.122(B)]

3.7 Neutral current (fundamental)

$I_N = \sqrt{(I_A^2 + I_B^2 + I_C^2 - I_A \cdot I_B - I_B \cdot I_C - I_C \cdot I_A)}$ (L-N loads only)
Balanced linear L-N load → $I_N \approx 0$. L-L and 3Ø loads contribute nothing.
Feeder neutral may be reduced to the max unbalanced load [220.61]
(no reduction for nonlinear loads – 220.61(C))

3.8 Harmonics — neutral and transformer

Triplen harmonics (3rd, 9th ...) ADD in the neutral:
 $I_{N,3} = \Sigma$ over phases of $I_{3,phase}$ balanced: $I_{N,3} = 3 \times I_{3,phase}$
Total neutral: $I_N = \sqrt{(I_{N,1}^2 + I_{N,3}^2 + I_{N,9}^2 + \dots)}$
Balanced single-3rd approximation:
 $I_N / I_{phase,rms} = 3 \cdot h_3 / \sqrt{(1 + h_3^2)}$ $h_3 = I_3 / I_1$
 $h_3 = 0.33 \rightarrow 0.94$ $h_3 = 0.50 \rightarrow 1.34$ $h_3 = 0.70 \rightarrow 1.72$ (→ 200% neutral)
Phase rms: $I_{rms} = I_1 \times \sqrt{(1 + THD_I^2)}$
Transformer K-factor: $K = \Sigma (I_h / I_R)^2 \times h^2$ (IEEE C57.110 / UL 1561)
A standard 75 kVA unit feeding heavy nonlinear load may need derating – ask the manufacturer or apply C57.110 capability equation.

3.9 Grounding and bonding conductor sizing

EGC: Table 250.122 by OCP rating (see Table 4.2)
Parallel runs: full-size EGC in each raceway [250.122(F)]
GEC (SDS): Table 250.66 by largest ungrounded conductor [250.30(A)(5)]
System bonding jumper / supply-side bonding jumper:
 Table 250.102(C)(1); above 1100 kcmil Cu → 12.5% of phase CM
 [250.28(D), 250.102(C)]
IG conductor: size as EGC [250.146(D)]

3.10 Raceway and box sizing

Fill: Σ conductor areas (Ch. 9 Table 5/8) $\leq 40\%$ of raceway area (Table 4) [Ch. 9 Table 1]
Conductors ≥ 4 AWG – pull/junction boxes: [314.28(A)]
 straight pull: box length $\geq 8 \times$ largest trade size (3" → 24 in)
 angle / U pull: distance $\geq 6 \times$ largest trade size + Σ other trade sizes on same wall
 (two 3" + one 2" on one wall → $6 \times 3 + 3 + 2 = 23$ in)
RMC bends between pull points $\leq 360^\circ$ total [344.26]
RMC support: within 3 ft of box, ≤ 10 ft; straight runs per Table 344.30(B)(2)
(3" RMC → 20 ft with threaded couplings)

3.11 Transformer protection and secondary conductors

$I_{pri} = kVA \times 1000 / (\sqrt{3} \times V_{pri})$ $I_{sec} = 75,000 / 360.3 = 208$ A
Table 450.3(B), primary + secondary protection:
 primary $\leq 250\%$ of I_{pri} , secondary $\leq 125\%$ of $I_{sec} = 260$ A → next std. 300 A (Note 1)
 (existing 250 AT main is within this limit)
Secondary conductors without OCP at the transformer must meet 240.21(C):
 (C)(2) ≤ 10 ft or (C)(6) ≤ 25 ft, ampacity $\geq 1/3 \times$ primary OCP $\times (V_{pri}/V_{sec})$,
 terminate in a single OCP that limits load to conductor ampacity.
Transformer loading %: $S_{total} / 75$ kVA $\times 100$

3.12 Motor and HVAC branch circuits

Conductor $\geq 1.25 \times \text{FLC}$ (single motor) [430.22]
Feeder with several motors: $1.25 \times \text{largest FLC} + \Sigma \text{ other FLC} + \text{other loads}$ [430.24]
Inverse-time breaker $\leq 2.50 \times \text{FLC}$ (round up to std. size) [430.52, Table 430.52]
Hermetic A/C: conductor $\geq \text{MCA}$; OCP $\leq \text{MOP}$ [440.32, 440.22]

3.13 PDR (UPS) and battery

Output: $I_{\text{out}} = P_{\text{out, rated}} / (360.3 \times \text{PF}_{\text{out}})$ 50 kW @ PF 1.0 $\rightarrow$ 139 A
Output OCP $\approx 1.25 \times I_{\text{out}} \rightarrow 173.6 \rightarrow 175$ A (confirm with PDR submittal)
Input: $I_{\text{in}} \approx (P_{\text{load}} / \eta + P_{\text{recharge}}) / (360.3 \times \text{PF}_{\text{in}})$
Critical load check: $\Sigma P_{\text{crit}} \leq P_{\text{rated}}$ (design $\leq 80\%$) and $\Sigma S_{\text{crit}} \leq S_{\text{rated}}$
DC current (worst case, end of discharge): $I_{\text{dc}} = P_{\text{load}} / (\eta_{\text{inv}} \times V_{\text{dc, min}})$
Runtime (approx.): $t_{\text{h}} = E_{\text{kWh}} \times \text{DoD} \times \eta_{\text{inv}} / P_{\text{load, kW}}$
ESS circuit sizing [706.30]

3.14 Short-circuit — point-to-point method

Transformer secondary, infinite primary:
 $I_{\text{sc}, 0} = I_{\text{FLA}} \times 100 / \%Z$ (75 kVA: 3%Z $\rightarrow$ 6.94 kA, 5%Z $\rightarrow$ 4.16 kA)
(more exact: include utility source impedance)
At a downstream point (3Ø fault):
 $f = \sqrt{3} \times L \times I_{\text{sc, prev}} \times Z_{\text{ft}} / (n \times E_{\text{LL}})$ $Z_{\text{ft}} = |Z|$ per ft from Ch. 9 Table 9
(Bussmann form: $f = \sqrt{3} \cdot L \cdot I / (C \cdot n \cdot E)$, $C = 1 / Z_{\text{ft}}$)
 $M = 1 / (1 + f)$ $I_{\text{sc, next}} = I_{\text{sc, prev}} \times M$
L-N fault: $f = 2 \times L \times I_{\text{LN}} / (C \times n \times E_{\text{LN}})$, with $I_{\text{LN}} \approx 1.5 \times I_{\text{3Ø}}$ at the X0 terminal
Motor contribution: $\approx 4 \times \Sigma \text{ motor FLC}$ (add at the point of interest)
PDR (inverter) contribution: typically 1.5–2 $\times$ rated, time-limited – use manufacturer data
Equipment rating $\geq$ available fault current [110.9, 110.10]

Example. 75 kVA, 3% Z $\rightarrow I_{\text{sc}, 0} = 6,940$ A. 50 ft of one set 250 kcmil Cu in steel conduit: $|Z| = \sqrt{(0.054^2 + 0.052^2)} = 0.075 \Omega/1000 \text{ ft} \rightarrow f = 0.217$, $M = 0.822$, I_{sc} at the IFS $\approx$ **5,704 A** (before HEMP filter impedance, which lowers it further). 42 kAIC breakers are well above this; confirm the busway and plug-in unit SCCR.

3.15 Test acceptance values (commissioning)

Insulation resistance (600 V class): 1000 VDC test, 1 min
minimum per ANSI/NETA ATS Table 100.1 (commonly 100 MΩ for 600 V cable/bus)
Polarization index: $\text{PI} = R_{10\text{min}} / R_{1\text{min}} \geq 2.0$ generally acceptable
Bolted joints: torque per manufacturer; micro-ohm across joints, compare similar joints
(NETA flags deviation $> 50\%$ from lowest value)
Infrared (under load): ΔT between similar components 1–3 °C possible deficiency,
4–15 °C probable deficiency, > 15 °C major (NETA ATS/MTS Table 100.18)